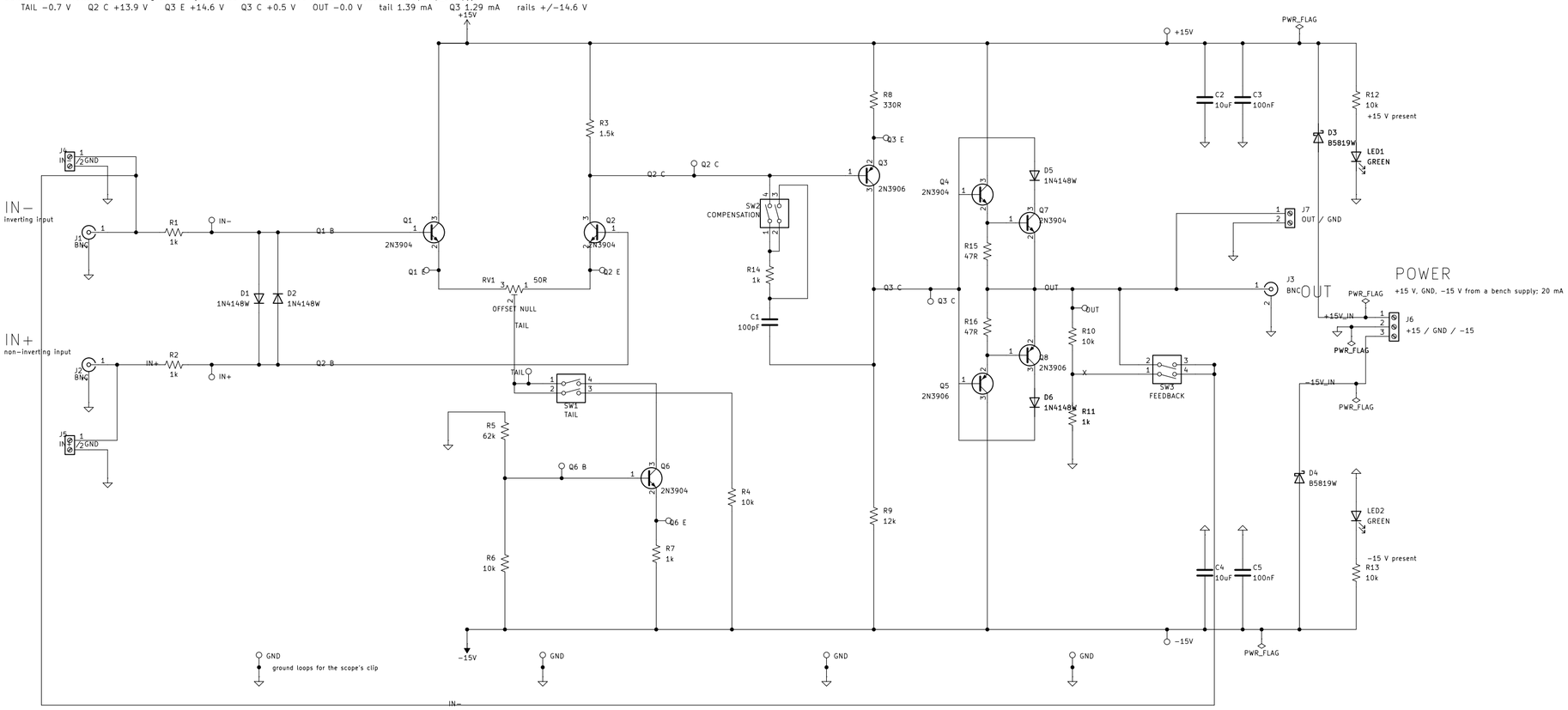
DISCRETE TRANSISTOR OP-AMP

The op-amp of Lab 5L (Learning the Art of Electronics, Hayes & Horowitz, 2016), built from the NPN and
PNP of Chapter 4 as six SOT-23 transistors, on a board whose silkscreen is this drawing.
Three stages, as the book's Fig. 5L.11 and 5N.32:

    difference   Q1 Q2, 1.5k collector load on the + side, tail from RV1 through SW1   x11
    gain   Q3, PNP common emitter: 12k / (330 + re) = 30; the push-pull's dead zone takes a third of it   x12
    follower   Q4 Q5 push-pull, unbiased, so crossover distortion shows until feedback hides it   x1

Open-loop gain at DC about 138 (the book says 'approximately 1000' for its CA3096 version, before the
dead zone). As a follower it settles in 0.6 us with 51 deg of phase margin; the x11 amplifier
has a bandwidth of 422 kHz and a gain of 10.2, 7 % low, as the book predicts for so little loop gain.

Quiescent as a follower with IN+ grounded (simulated with the 2N3904/2N3906 models, sim/opamp_sim.py):
    TAIL -0.7 V   Q2 C +13.9 V   Q3 E +14.6 V   Q3 C +0.5 V   OUT -0.0 V   tail 1.39 mA   Q3 1.29 mA   rails +/-14.6 V



INPUTS
Each input has a BNC and a screw terminal in
parallel; +/-12 V on either. For a common-mode
test, tee one generator into both jacks.

INPUT PROTECTION (R1, R2, D1, D2: not in the book)
1k in series with each input, and two 1N4148
back to back between the bases: the bases never
differ by more than 0.7 V, so no base-emitter
junction sees its 6 V reverse breakdown when one
input is driven hard and the other is not.
A 1 M probe or a 50 ohm generator never notices
the 1k; Ib (about 5 uA) drops 5 mV across it.

TAIL (SW1)
1 ON: Q6, the 'simple sink' of Fig. 5L.10, 1.4 mA
2 ON: 10k to -15 V, 1.4 mA (Fig. 5L.1)
The sink raises the tail's impedance from 10k to
about 1 M; stage 1's common-mode gain falls from
0.073 to 0.0004. Both OFF: no tail current, dead.
Normal: 1 ON, 2 OFF.

OFFSET NULL (RV1: not in the book)
The two halves of the 50 ohm trimmer are the pair's
emitter resistors, 25 ohm each at centre. Turning it
unbalances the pair by up to +/-33 mV at the input:
enough to centre the output in open loop and to
cancel the pair's Vbe mismatch in closed loop.
It costs half of stage 1's gain (11 instead of 20).

COMPENSATION (SW2)
1 ON: 100 pF from Q3's collector to its base, Miller-
multiplied by the stage's gain of 30 into a 3 nF
low-pass at C2. The book's Fig. 5L.17.
2 ON: shorts the 1k, leaving the bare 100 pF.

The 1k is not in the book. A bare Miller cap feeds
the signal FORWARD above 4 MHz, the loop's sign
flips, and a follower oscillates: the book's 'risky
business'. With the 1k the follower has 51 deg of
phase margin and settles in 0.6 us; the x11 amp is
stable either way. Both OFF: no compensation,
and the x11 amplifier rings.
Normal: 1 ON, 2 OFF.

FEEDBACK (SW3)
1 ON: X to IN-, gain 11 (Fig. 5L.16)
2 ON: OUT to IN-, a follower (Fig. 5L.18)
Both OFF: nothing; wire your own network between
the OUT, IN- and GND terminals. The feedback goes
to the IN- terminal, so R1 and the clamp still guard
Q1's base. The 10k + 1k are the book's load and its
x11 divider in one: X = OUT / 11.

OUTPUT
-14 to +13 V into the 11k: 10 mA max.
Q7, Q8, D5, D6 and the 47 ohm resistors are not in the
book: they limit the output to about 20 mA, as a real
op-amp does, so a short to a rail or ground is survived.

POWER
+15 V, GND, -15 V from a bench supply, 20 mA.
Series Schottkys: a supply connected backwards
does nothing instead of forward-biasing every
transistor. The rails run 0.35 V below the
terminals: +/-14.6 V on the board. Everything
in the book's circuit is referred to those rails,
and nothing in it minds. A green LED per rail
says the supply is on and the right way round.

PROBING
A 2.54 mm pin on every node named on the sheet;
a scope hook grabs it. Three pairs of ground
pins for the clip.

MH1 MH2 MH3 MH4

M3 mounting holes, one per corner

Open source, CERN-OHL-P-2.0 -- github.com/gallicchio/op-amp-pcb
difference x11, gain x12, follower x1: A = 138 open loop (crossover included)
2N3904 / 2N3906 (MMBT, SOT-23): the transistors of Chapter 4
Circuit: Hayes & Horowitz, Learning the Art of Electronics (2016), Lab 5L | Board: Jason Gallicchio
Sheet: /
File: opamp.kicad_sch

Title: Discrete Transistor Op-Amp: the Lab 5L op-amp of Learning the Art of

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